

BADIN, Pakistan

WMO: 417850

Lat: 24.633N

Lon: 68.900E

Elev: 11

StdP: 101.19

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	10.7	-0.6	3.6	21.5	1.4	4.2	20.1	5.4	22.4	4.7	21.8	0.3	50	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.2	40.9	25.0	39.6	24.9	38.5	25.2	28.5	35.0	28.1	34.5	27.7	33.9	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	31.2	26.7	22.3	30.9	26.3	21.8	30.7	92.7	35.5	90.8	34.6	89.2	34.0	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.5	7.3	43.5	3.2	1.1	5.1	44.3	3.2	45.0	1.4	45.6	-0.9	46.4		
			5.7	29.7	1.9	0.6	4.4	30.2	3.3	30.6	2.2	30.9	0.9	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	18.3	21.6	26.7	30.8	33.0	33.1	31.4	30.0	30.0	29.4	24.5	19.6
	DBStd	5.34	2.26	3.04	2.58	1.66	1.22	1.26	1.38	1.08	1.15	1.43	2.50	2.48
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	52	30	7	0	0	0	0	0	0	0	0	0	14
	CDD10.0	6351	257	324	518	625	714	694	664	621	600	601	436	297
	CDD18.3	3361	28	98	260	375	456	444	406	362	350	343	186	53
	CDH23.3	43635	256	882	2849	4971	6674	6692	5772	4694	4523	4192	1716	414
	CDH26.7	24017	30	301	1526	3003	4220	4295	3302	2241	2182	2139	697	82
Wind	WSAvg	2.1	0.9	1.1	1.6	2.5	4.0	3.9	3.6	2.9	2.3	1.2	0.7	0.8
Precipitation	PrecAvg	247	2	7	1	2	6	10	71	83	53	10	2	1
	PrecMax	902	14	61	5	18	161	64	301	355	350	116	36	8
	PrecMin	36	0	0	0	0	0	0	3	1	4	0	0	0
	PrecStd	193	3	14	1	4	30	15	83	99	94	27	7	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	34.2	39.3	41.8	42.9	42.3	39.0	35.8	37.3	37.7	34.5	30.4
		MCWB	17.3	19.6	21.8	23.2	24.7	26.4	27.6	28.0	27.2	25.3	21.8	19.4
	2%	DB	27.4	32.2	37.2	40.1	41.2	40.0	37.1	34.7	35.7	36.2	33.0	28.8
		MCWB	16.6	19.3	21.6	23.3	25.2	27.0	27.4	27.2	27.2	24.5	21.6	18.4
	5%	DB	26.1	30.2	35.4	39.0	39.8	38.5	36.0	33.7	34.4	35.1	32.0	27.4
		MCWB	16.3	18.7	20.7	23.1	25.2	27.1	27.3	26.8	26.8	24.3	21.4	17.7
	10%	DB	24.6	28.4	33.8	37.6	38.5	37.4	35.0	33.0	33.4	34.3	30.7	25.7
		MCWB	15.3	17.6	20.2	22.9	25.4	27.1	27.1	26.6	26.4	24.2	21.0	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	22.3	24.4	26.5	28.5	28.9	29.0	28.8	28.8	27.6	24.8	21.7
		MCDB	26.6	30.1	34.4	36.9	37.2	37.3	35.3	34.1	34.4	34.4	30.9	27.9
	2%	WB	18.7	21.2	23.2	25.6	27.7	28.2	28.2	28.1	28.2	26.7	23.7	20.2
		MCDB	24.7	28.7	33.2	35.4	35.9	36.0	34.3	32.8	33.4	32.9	29.6	26.0
	5%	WB	17.4	20.2	22.4	25.0	27.3	27.9	27.8	27.6	27.7	26.2	22.9	19.0
		MCDB	23.7	27.0	31.7	33.6	35.0	35.4	33.7	32.1	32.6	32.3	28.8	25.2
	10%	WB	16.2	19.2	21.6	24.4	26.8	27.6	27.5	27.2	27.1	25.7	22.1	17.9
		MCDB	23.0	26.2	30.2	33.0	34.1	34.7	33.1	31.5	31.8	31.6	28.2	24.4
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.1	13.8	14.0	12.1	9.2	6.9	6.0	7.2	10.2	11.8	11.9
		MCDBR	14.0	14.7	16.0	15.7	14.1	11.2	8.6	7.2	9.0	11.3	12.3	13.1
		MCWBR	5.4	4.2	3.6	3.6	2.8	2.1	1.7	1.9	2.6	2.6	3.5	4.8
	5% WB	MCDBR	12.4	12.8	14.3	14.3	11.6	9.4	7.6	6.7	7.7	9.4	10.6	11.5
		MCWBR	6.1	4.4	4.0	3.7	2.4	2.0	2.1	2.4	2.7	2.5	3.7	5.0
Clear-Sky Solar Irradiance	taub	0.443	0.469	0.530	0.584	0.642	0.724	0.778	0.709	0.610	0.558	0.539	0.487	
	taud	2.020	1.987	1.847	1.767	1.697	1.587	1.534	1.648	1.810	1.826	1.818	1.909	
	Ebn at Noon	795	807	778	746	701	642	607	649	705	715	693	727	
	Edh at Noon	154	171	206	228	244	271	285	254	212	198	188	167	
All-Sky Solar Radiation	RadAvg	4.30	5.17	6.10	6.79	7.02	6.26	5.06	5.00	5.55	5.43	4.45	4.04	
	RadStd	0.15	0.17	0.20	0.20	0.17	0.39	0.35	0.29	0.29	0.21	0.17	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page